2.03 Graph Sketching: $\sin x$, $\cos x$, $\tan x$ in Radians

Worked Solutions 1–25

1 a)

$y = 3\sin x$.

1 b)

$y = \frac{4}{5}\sin x$.

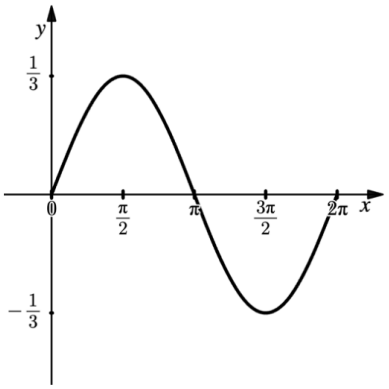
2 a)

$y = \frac{5}{2}\cos x$.

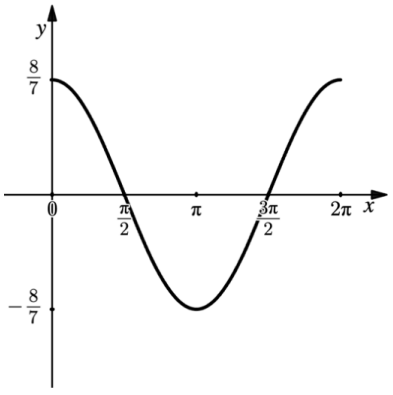
2 b)

$y = \frac{9}{4}\cos x$.

3 a)



3 b)



4 a)

$y = \sin 2x$.

4 b)

$y = \sin 3x.$

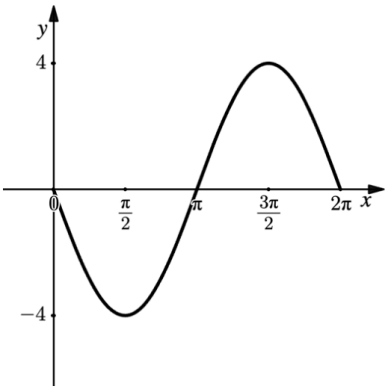
4 c)

$y = \sin \frac{5}{6}x.$

4 d)

$y = \sin \frac{3}{5}x.$

5)



6 a)

$y = \cos \frac{5}{2}x.$

6 b)

$y = \cos \frac{3}{4}x.$

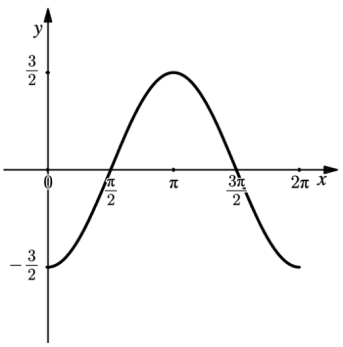
6 c)

$y = \cos \frac{6}{5}x.$

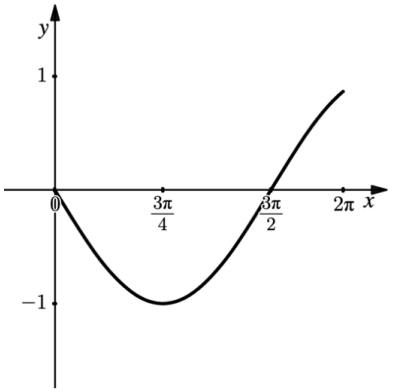
6 d)

$y = \cos \frac{5}{8}x.$

7)



8)



9 a)

$y = \tan 2x$.

9 b)

$y = \tan \frac{3}{2}x$.

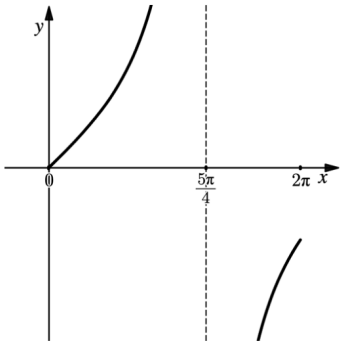
9 c)

$y = \tan \frac{4}{3}x$.

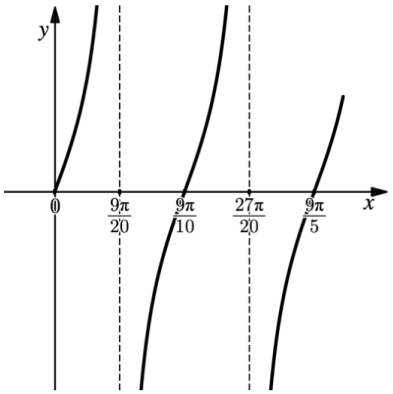
9 d)

$y = \tan \frac{8}{9}x$.

10 a)



10 b)



11 a)

$y = 3 + \sin x$.

11 b)

$y = -4 + \cos x$.

11 c)

$y = -1 + \sin x$.

11 d)

$y = 2 + \cos x$.

12 a)

$y = 3 + 2\sin x$.

12 b)

$y = -2 + 2\cos x$.

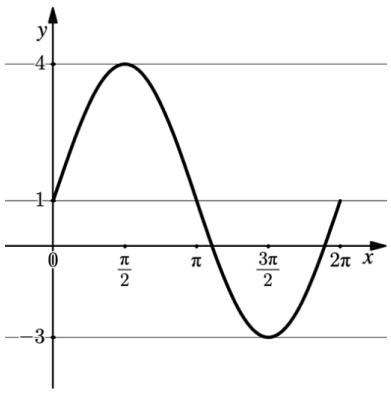
12 c)

$y = 1 + 4\cos x$.

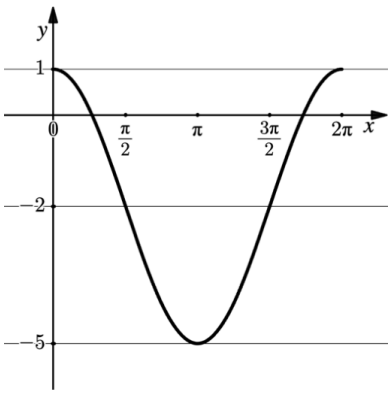
12 d)

$y = 2 - 3\sin x$.

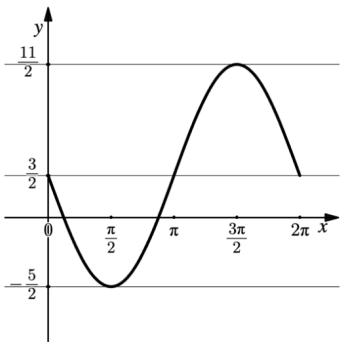
13 a)



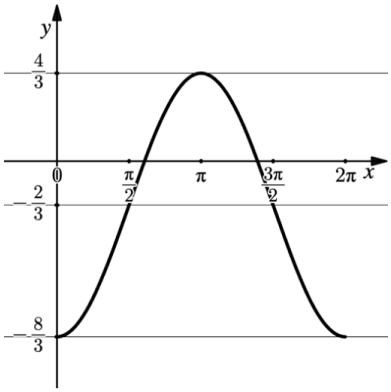
13 b)



13 c)



13 d)



14 a)

$y = \sin(x - \frac{\pi}{2})$.

14 b)

$y = \sin(x + \frac{\pi}{3})$.

15 a)

$y = \cos(x + \frac{2\pi}{3})$.

15 b)

$y = \cos(x - \frac{3\pi}{5})$.

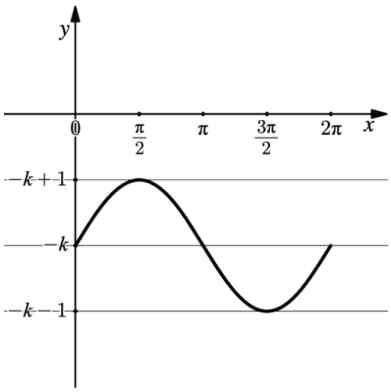
16 a)

$y = \tan(x + \frac{\pi}{6})$.

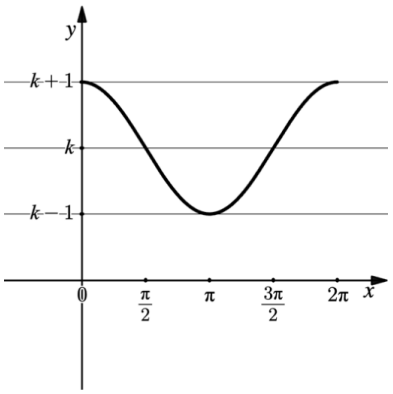
16 b)

$y = \tan(x - \frac{2\pi}{3})$.

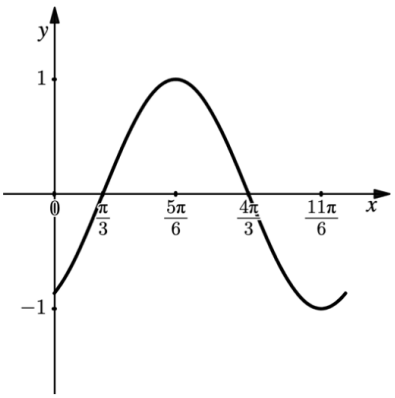
17 a)



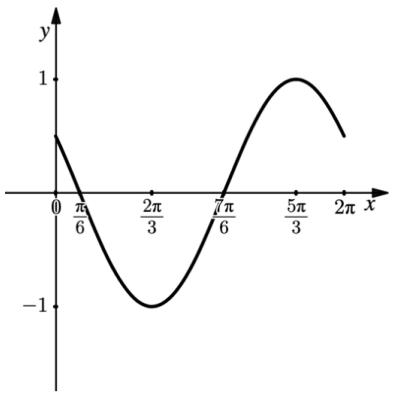
17 b)


y
k+1
k
k−1
0
π/2
π
3π/2
2π
x

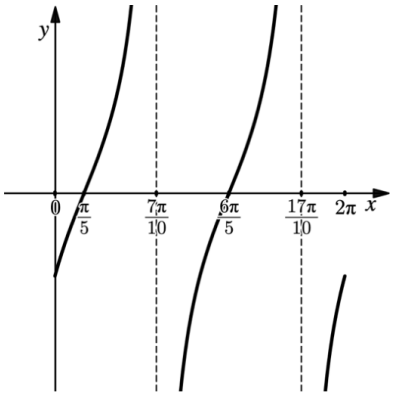
18 a)


y
1
−1
0
π/3
5π/6
4π/3
11π/6
x

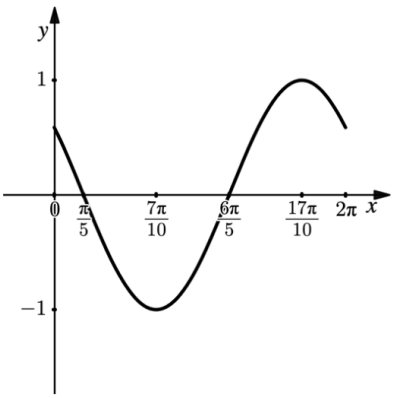
18 b)


y
1
−1
0
π/6
2π/3
7π/6
5π/3
2π
x

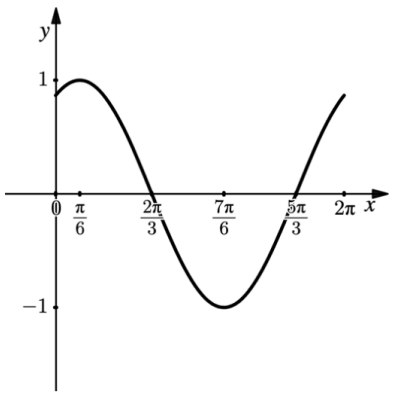
18 c)


y
0
π/5
7π/10
6π/5
17π/10
2π
x

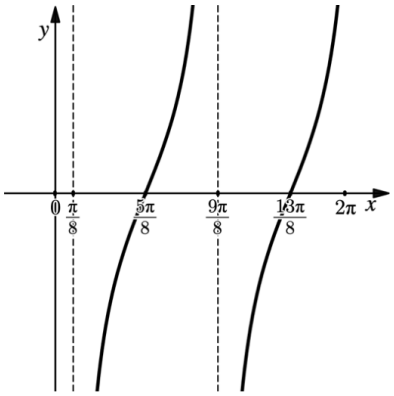
18 d)


y
1
0
π/5
7π/10
6π/5
17π/10
2π
x
−1

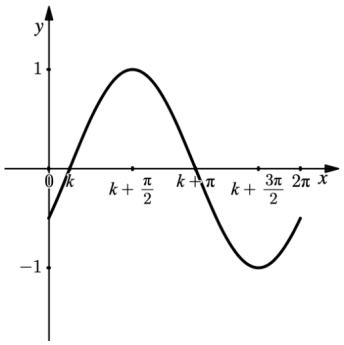
18 e)


y
1
0
π/6
2π/3
7π/6
5π/3
2π
x
−1

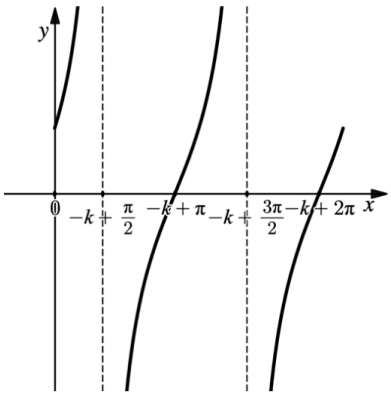
18 f)



19 a)



19 b)



20 a)

$y = \sin(2x - \frac{\pi}{3})$.

20 b)

$y = \sin(\frac{1}{2}x + \frac{\pi}{4})$.

20 c)

$y = \sin(\frac{2}{3}x - \frac{\pi}{5})$.

21 a)

$y = \cos(\frac{1}{2}x - \frac{\pi}{8})$.

21 b)

$y = \cos(\frac{5}{3}x + \frac{\pi}{5})$.

21 c)

$y = \cos(\frac{5}{7}x - \frac{5\pi}{9})$.

22 a)

$y = \tan(\frac{5}{3}x + \frac{\pi}{3})$.

22 b)

$y = \tan(\frac{7}{8}\pi - \frac{\pi}{4})$.

22 c)

$y = \tan(\frac{4}{9}x + \frac{\pi}{6})$.

23 a)

$y = 2 + 3\sin(x + \frac{\pi}{8})$.

23 b)

$y = -3 + \sin(\frac{3}{2}x - \frac{\pi}{10})$.

23 c)

$y = -1 + 4\sin(\frac{4}{5}x + \frac{\pi}{4})$.

23 d)

$y = 2 - 3\sin(\frac{5}{4}x - \frac{\pi}{3})$.

24 a)

$y = 4\cos(\frac{3}{4}x - \frac{\pi}{4})$.

24 b)

$y = -2 + 2\cos(\frac{4}{5}x + \frac{2\pi}{5})$.

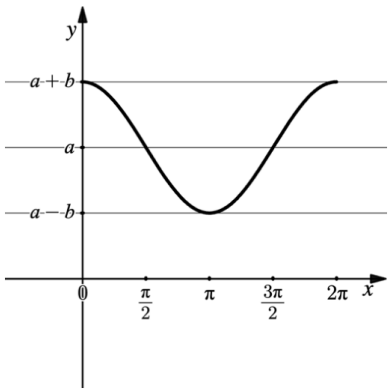
24 c)

$y = -1 - 4\cos(\frac{7}{9}x - \frac{\pi}{6})$.

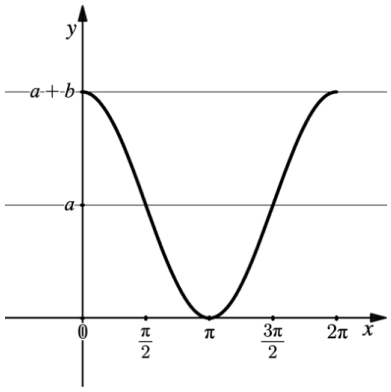
24 d)

$y = 2 - 3\cos(\frac{7}{10}x + \frac{4\pi}{5})$.

25 a)



25 b)



25 c)

